Discrete Mathematics Cheatsheet

malloc.nvim

1 Propositional Logic

1.1 Propositions

A proposition is a declarative statement that has exactly one truth value: either True (T) or False (F).

1.2 Logical Connectives

1.2.1 Logical Operators

Symbol	Name	Meaning
$\neg p$	Negation	NOT p
$p \wedge q$	Conjunction	p AND q
$p \vee q$	Disjunction	p OR q
$p \to q$	Implication	IF p THEN q
$p \leftrightarrow q$	Biconditional	IFF
$p \oplus q$	XOR	Exactly one

1.2.2 Truth Table

p	q	$\neg p$	$p \wedge q$	$p \vee q$	$p \to q$	$p \leftrightarrow q$	$p \oplus q$
T	T	F	T	T	T	T	F
T	F	F	F	T	F	F	T
F	T	T	F	T	T	F	T
F	F	T	F	F	T	T	F

1.3 Propositional Equivalences

1.3.1 Logical Equivalences

Law	Equivalence
Identity	$p \wedge T \equiv p, \quad p \vee F \equiv p$
Domination	$p \vee T \equiv T, \quad p \wedge F \equiv F$
Idempotent	$p \wedge p \equiv p, \quad p \vee p \equiv p$
Double Negation	$\neg\neg p \equiv p$
Commutative	$p \wedge q \equiv q \wedge p$ $p \vee q \equiv q \vee p$
Associative	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$ $(p \vee q) \vee r \equiv p \vee (q \vee r)$
Distributive	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$ $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
De Morgan	$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$
Negation	$p \vee \neg p \equiv T, \quad p \wedge \neg p \equiv F$
Absorption	$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$

Law	Equivalence
Conditional	$p \to q \equiv \neg p \vee q$
Converse	$q \to p$
Inverse	$\neg p \to \neg q$
Contrapositive	$\neg q \to \neg p$
Negation of Conditional	$\neg(p \to q) \equiv p \wedge \neg q$

1.3.2 Logical Equivalences Involving Conditional and Biconditional Statements

Law	Equivalence
Conditional Identity	$p \to q \equiv \neg p \vee q$
Contrapositive	$p \to q \equiv \neg q \to \neg p$
Negation of Conditional	$\neg(p \to q) \equiv p \wedge \neg q$
Biconditional Identity	$p \leftrightarrow q \equiv (p \to q) \wedge (q \to p)$
Biconditional (Expanded)	$p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$
Negation of Biconditional	$\neg(p \leftrightarrow q) \equiv (p \wedge \neg q) \vee (\neg p \wedge q)$

1.3.3 Translation of Logical Equivalences

English Phrase	Logical Form
not p	$\neg p$
p and q	$p \wedge q$
p or q	$p \vee q$
Either p or q, but not both	$p \oplus q$
If p, then q	$p \to q$
p only if q	$p \to q$
p if q	$q \to p$
p if and only if q	$p \leftrightarrow q$
p is sufficient for q	$p \to q$
p is necessary for q	$q \to p$
Neither p nor q	$\neg p \wedge \neg q$
Not both p and q	$\neg(p \wedge q)$
At least one of p, q	$p \vee q$
Both p and q are false	$\neg p \wedge \neg q$

1.4 Predicates and Quantifiers

A predicate $P(x)$ is a statement involving a variable that becomes a proposition when a value is assigned to the variable.

The domain of discourse is the set of values a variable is allowed to take.

Symbol	Name	Meaning
$\forall x\, P(x)$	Universal	$P(x)$ holds for every x
$\exists x\, P(x)$	Existential	$P(x)$ holds for at least one x
$\exists! x\, P(x)$	Unique existence	$P(x)$ holds for exactly one x

Example: Let $P(x)$ mean "$x > 0$" over $\mathbb{Z}$. Then $\forall x\, P(x)$ is false, while $\exists x\, P(x)$ is true.

1.4.1 De Morgan's Laws for Quantifiers

Law	Equivalence
Universal Negation	$\neg\forall x\, P(x) \equiv \exists x\, \neg P(x)$
Existential Negation	$\neg\exists x\, P(x) \equiv \forall x\, \neg P(x)$
Negated Universal	$\neg(\forall x\, P(x)) \equiv \exists x\, \neg P(x)$
Negated Existential	$\neg(\exists x\, P(x)) \equiv \forall x\, \neg P(x)$

1.4.2 Multiple Quantifiers

Statement	Meaning
$\forall x \forall y\, P(x, y)$	P holds for every pair (x, y)
$\exists x \exists y\, P(x, y)$	P holds for at least one pair
$\forall x \exists y\, P(x, y)$	For every x, some y works
$\exists y \forall x\, P(x, y)$	One y works for every x

1.4.3 Translation into Predicate Logic

English	Logical Form
Every x has property P	$\forall x\, P(x)$
Some x has property P	$\exists x\, P(x)$
No x has property P	$\forall x\, \neg P(x)$
Not every x has property P	$\exists x\, \neg P(x)$
Everyone is P and Q	$\forall x\, (P(x) \wedge Q(x))$
Everyone is P or Q	$\forall x\, (P(x) \vee Q(x))$
Someone is P and Q	$\exists x\, (P(x) \wedge Q(x))$
Someone is P or Q	$\exists x\, (P(x) \vee Q(x))$
Every P is Q	$\forall x\, (P(x) \to Q(x))$
Some P is Q	$\exists x\, (P(x) \wedge Q(x))$
No P is Q	$\forall x\, (P(x) \to \neg Q(x))$
Some P is not Q	$\exists x\, (P(x) \wedge \neg Q(x))$

1.5 Rules of Inference

Rule	Form
Modus Ponens (MP)	$\dfrac{p \quad p \to q}{\therefore q}$
Modus Tollens (MT)	$\dfrac{p \to q \quad \neg q}{\therefore \neg p}$
Hypothetical Syllogism (HS)	$\dfrac{p \to q \quad q \to r}{\therefore p \to r}$
Disjunctive Syllogism (DS)	$\dfrac{p \vee q \quad \neg p}{\therefore q}$
Addition	$\dfrac{p}{\therefore p \vee q}$
Simplification	$\dfrac{p \wedge q}{\therefore p}$
Conjunction	$\dfrac{p \quad q}{\therefore p \wedge q}$
Resolution	$\dfrac{p \vee q \quad \neg p \vee r}{\therefore q \vee r}$
Universal Instantiation (UI)	$\dfrac{\forall x\, P(x)}{\therefore P(c)}$
Existential Instantiation (EI)	$\dfrac{\exists x\, P(x)}{\therefore P(c)}$
Universal Generalization (UG)	$\dfrac{P(c),\ c \text{ arbitrary}}{\therefore \forall x\, P(x)}$
Existential Generalization (EG)	$\dfrac{P(c)}{\therefore \exists x\, P(x)}$

1.6 Introduction to Proofs

1.6.1 Choosing a Proof Method

Can p directly imply q? Direct Proof
Is $\neg q \to \neg p$ easier to prove? Contraposition
Is the hypothesis always false? Vacuous Proof
Is the conclusion always true? Trivial Proof
Does assuming the opposite lead to a contradiction? Proof by Contradiction
Can the statement be divided into exhaustive cases? Proof by Cases
Is the statement about all integers $n \geq n_0$? Mathematical Induction
Can one example disprove the claim? Counterexample